

Engineering Curves: Construction of Cycloid

Rolling Circle Method with Tangent & Normal

Gajavada Sanjeevkumar

Problem Statement

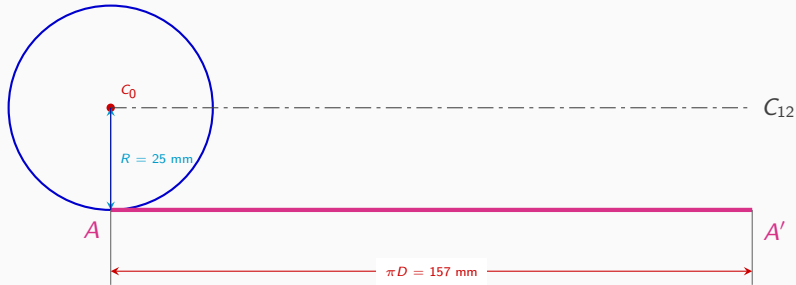
Question

Draw a **cycloid** generated by a circle of diameter $D = 50$ mm rolling along a straight base line for **one complete revolution** without slipping.

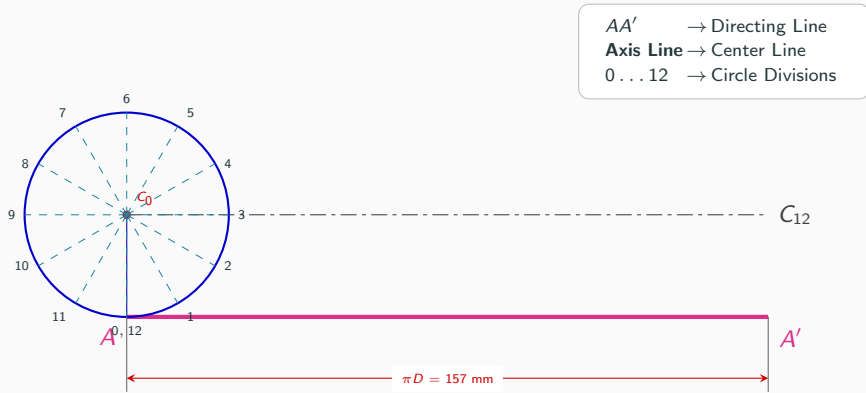
Construct a **tangent** and a **normal** to the cycloid at any point P located on the curve.

Step 1: Draw Generating Circle & Directing Line

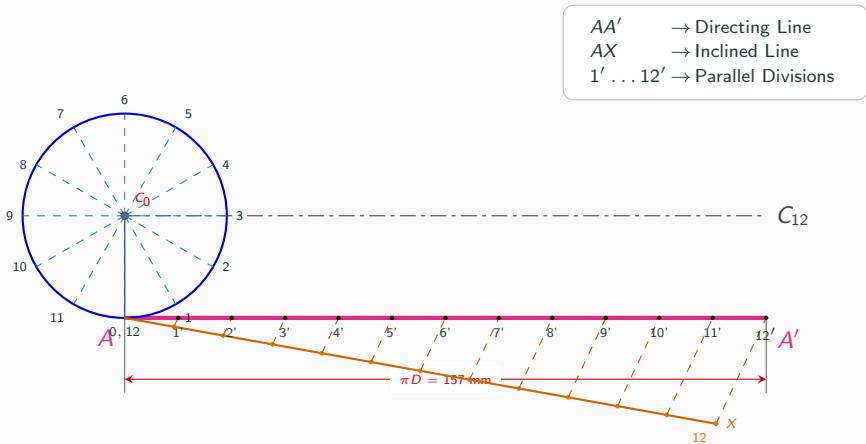
C_0 → Initial Center Point
 AA' → Directing Line
Axis → Center Line



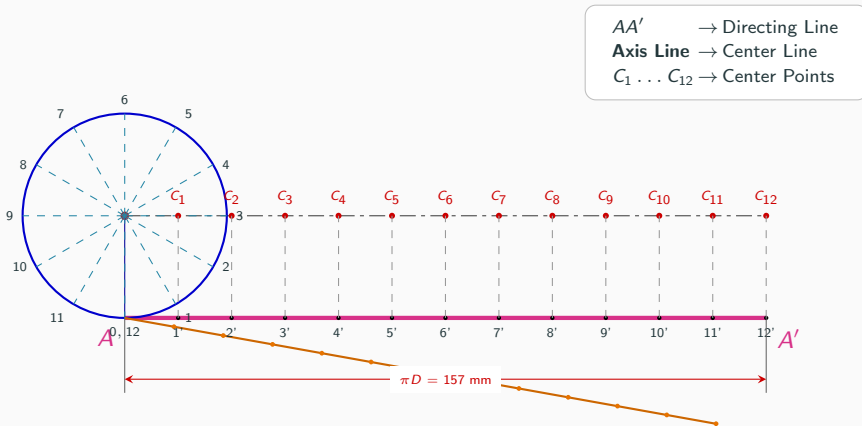
Step 2: Divide Generating Circle into 12 Equal Parts



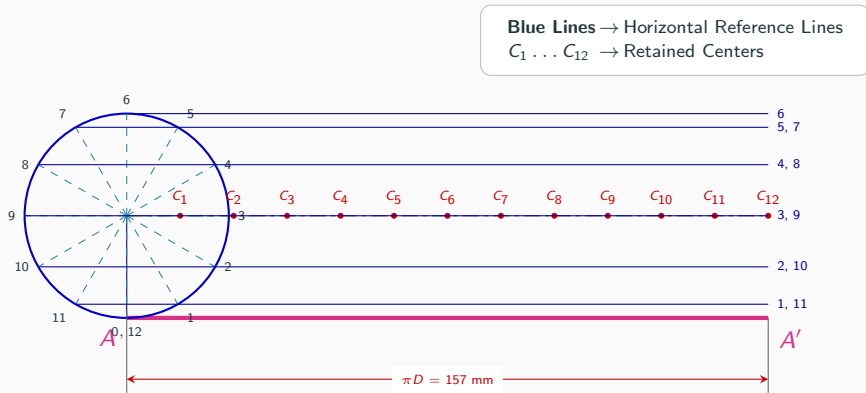
Step 3: Divide Directing Line AA' using Line Division Method



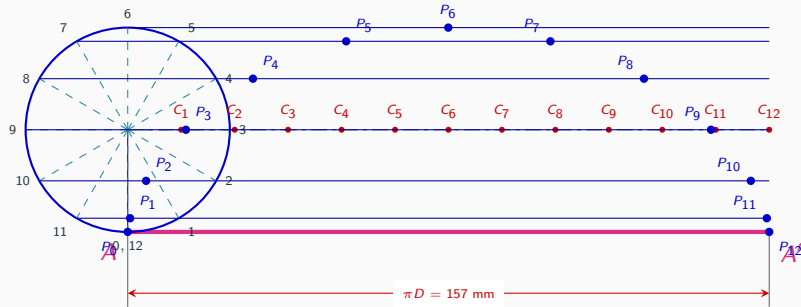
Step 4: Project Divisions to Locate Centers $C_1 \dots C_{12}$



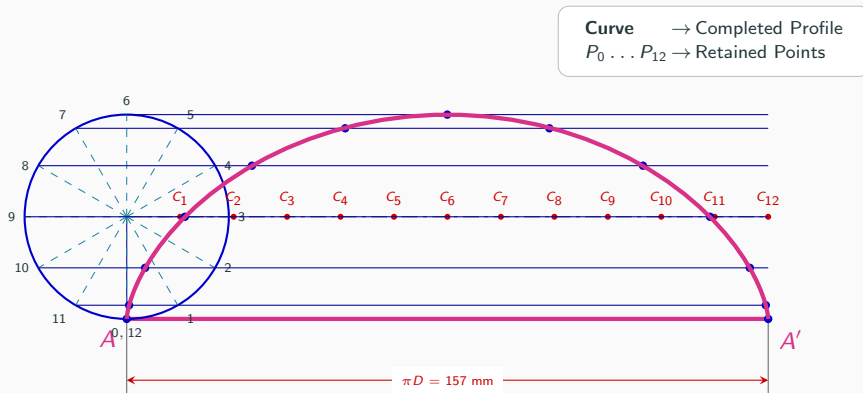
Step 5: Draw Horizontal Reference Lines



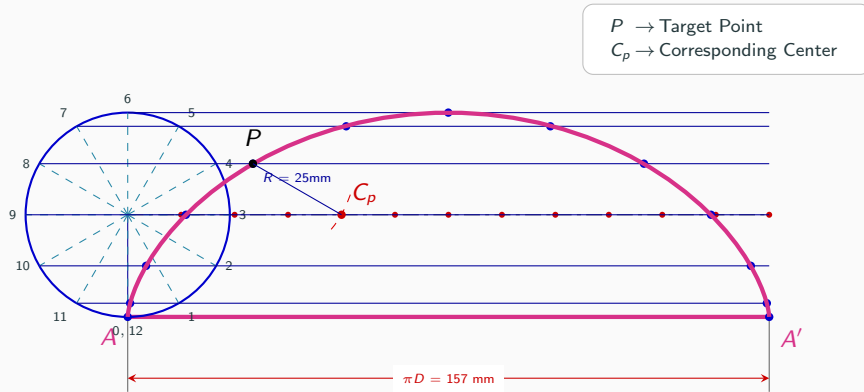
Step 6: Strike Arcs to Locate Points $P_1 \dots P_{12}$

$$\begin{aligned} C_1 \dots C_{12} &\rightarrow \text{Retained Centers} \\ P_1 \dots P_{12} &\rightarrow \text{Cycloid Locus Points} \end{aligned}$$


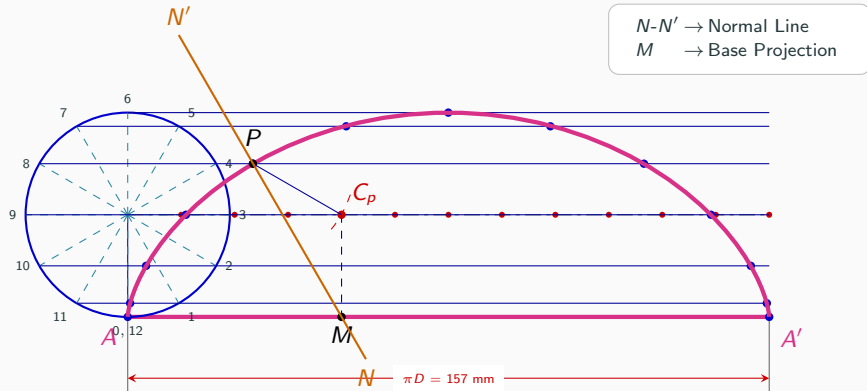
Step 7: Smoothly Join Points to Form Cycloid Curve



Step 8: Mark Point P & Locate Center C_p



Step 9: Construct Normal Line $N-N'$



Step 10: Construct Tangent Line $T-T'$ & Final Result

